
ISMB 2026 notification for paper 1058

From Abstracts - MLCSB: Machine Learning in Computational and Systems Biology <MLCSB-ISMB2026@easychair.org>

Date Tue 05/05/2026 21:57

To Trnovec, Lena <lena.trnovec@fri.uni-lj.si>

Dear Lena,

Your submission "Protein-informed single-cell embeddings preserve biological signal across large evolutionary distances" has been accepted for a poster at ISMB 2026 as part of the MLCSB COSI!

Please take a moment to review these details:

* Presenters of posters will receive a Confirmation of Participation (CoP) link that must be completed no later than June 5, 2026. This will be emailed directly to you from no-reply@iscb.org with the subject "ISMB 2026 Confirmation of Participation".

* The presenting author must register to attend the conference no later than June 5, 2026. Your registration ID is required to link your CoP with your talk or poster. Registration link:

<https://www.iscb.org/ismb2026/register>.

* ISCB is pleased to offer a limited number of fellowships to support registration and/or travel. Conference Fellowship Application Invitations are sent directly to eligible individuals after acceptance of a scientific submission to Proceedings, Abstracts, or Posters. More information regarding fellowships can be found at <https://www.iscb.org/ismb2026/general-info/conference-fellowships>

* All poster presenters (including those presenting in Washington, D.C.) must upload a PDF of their poster, but it is strongly encouraged to also upload a short 5–7 minute video. Failure to do so will result in not receiving a slot for the in-person sessions. All poster presenters must upload their poster details between June 9 and no later than 23:59 Anywhere on Earth on July 3. The Virtual Poster Theater, for virtual viewing of all posters, will open on July 9, 2026, for maximum exposure of each poster and to assist attendees in maximizing their time each day in the in-person Poster Hall.

Additional details are available at these links: <https://www.iscb.org/ismb2026/scientific-programme/posters>

<https://www.iscb.org/ismb2026/whats-happening/poster-sessions>

* Presenters should read all details and expectations provided at <https://www.iscb.org/iscb-policy-statements/presenter-information>

* For information about conference logistics, including FAQs and accommodations, please visit the ISMB 2026 website: <https://www.iscb.org/ismb2026/home> and review the subpages of General Information.

Thank you again for submitting your work to ISMB 2026, and we can't wait to see you July 12–16 Washington DC, USA!

David Knowles and Elana Fertig,
MLCSB Track Chairs.

SUBMISSION: 1058

TITLE: Protein-informed single-cell embeddings preserve biological signal across large evolutionary distances

----- REVIEW 1 -----

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TITLE: Protein-informed single-cell embeddings preserve biological signal across large evolutionary distances

----- Overall evaluation -----

The abstract benchmarks existing single-cell foundation models on non-mammalian organisms at large evolutionary distances, finding that ortholog mapping fails as distance increases. Using *C. elegans* and *D. discoideum* as test cases, they show that UCE, which represents genes via ESM-2 protein embeddings rather than gene names, retains far more biological signal zero-shot than competing methods. An interesting exception is *D. discoideum* pseudotime, where PCA exceeds the performance of UCE, likely because the slime mold's development follows a single dominant temporal axis that PCA's first component captures directly.

This seems like a clean use case and benchmark for the considered methods, and showcases clearly the cases where it would be useful to employ techniques beyond ortholog mapping such as UCE.

----- REVIEW 2 -----

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TITLE: Protein-informed single-cell embeddings preserve biological signal across large evolutionary distances

----- Overall evaluation -----

This is an excellent, timely benchmark that serves as a much-needed reality check for the field. The authors evaluate zero-shot cross-species transfer of single-cell foundation models on *D. discoideum* and *C. elegans*. They successfully pinpoint a critical bottleneck: models relying on human ortholog mapping (like Geneformer and scGPT) discard roughly two-thirds of the genes in these distant species. In contrast, UCE uses protein-sequence embeddings to retain over 97% of genes, leading to superior performance across 5 out of 6 curated annotation families. A major strength of this paper is its intellectual honesty. The authors explicitly highlight cases where simple, species-native baselines still outcompete massive foundation models, such as PCA achieving the best pseudotime agreement in *D. discoideum*. This objectivity makes the benchmark highly credible. The figures are clear, and the experimental design is solid.

----- REVIEW 3 -----

SUBMISSION: 1058

TITLE: Protein-informed single-cell embeddings preserve biological signal across large evolutionary distances

----- Overall evaluation -----

This abstract explores the question of cross-species single-cell analysis by evaluating whether protein-informed gene representations (such as ESM-2 used in UCE) preserve biological signals across large evolutionary distances. While the objective of finding species-agnostic gene representations is interesting, the current experimental design makes it difficult to definitively conclude that protein-based embeddings are the superior choice.

The primary concern regarding the claim of superiority is the gene-mapping discrepancy. If competing single-cell Foundation Models (scFMs) only retain a small fraction of genes after mapping, their performance drop may be a technical artifact of information loss rather than a failure of their underlying representation strategy. To support the claim, a more controlled comparison is needed where the feature space is more consistent across models.

The assertion that protein-based representations are the "best" choice for cross-species strategies

requires more rigorous support. The "zero-shot" performance is an interesting benchmark, but without addressing the aforementioned experimental flaws, the results feel preliminary rather than conclusive. Strengthening the experimental setup to ensure a "level playing field" for non-protein-based models would significantly increase the impact of this work.

----- REVIEW 4 -----

SUBMISSION: 1058

TITLE: Protein-informed single-cell embeddings preserve biological signal across large evolutionary distances

----- Overall evaluation -----

In this study, the authors present a zero-shot cross-species benchmark in two highly diverged organisms, *Dictyostelium discoideum* and *Caenorhabditis elegans*, comparing five single-cell embedding strategies. The motivation is that single-cell foundation models are mostly trained on mammalian data and evaluated in close phylogenetic relatives, and the authors check whether those models keep biologically meaningful signal when applied without retraining or fine-tuning to evolutionarily distant species. The results support the feasibility of cross-species transfer across large evolutionary distances and show that protein-based gene representation consistently outperforms ortholog mapping. The approach is well motivated and the evaluation is well-designed. One small point is that the benchmark covers only two organisms, testing on more organisms would make the conclusions stronger. Overall, I believe this work would be a valuable addition to the conference.

----- REVIEW 5 -----

SUBMISSION: 1058

TITLE: Protein-informed single-cell embeddings preserve biological signal across large evolutionary distances

----- Overall evaluation -----

This work sets to evaluate whether single-cell foundation models trained on mammalian data can transfer zero-shot to two out-of-distribution (evolutionarily remotely related) organisms: *C. elegans* and amoeba. The authors created a benchmark dataset of these 2 species and evaluated 3 recent pretrained models performance in typical single cell tasks like developmental trajectory inference, cell type prediction.

While the abstract is very well presented and formatted, some key technical details are missing. It seems that PCA and scVI were used as naïve benchmarks applied directly to the test data, so not applied in the framework of a generalization benchmark. It is not clear that this is an appropriate choice given the vast diversity of methods available in the field for these tasks.

Additionally, Geneformer and scGPT required mapping human orthologs because these two models define their vocabularies directly over human genes. Given the extra layer of mapping issues introduced here, is it still practical for amoeba or *C. elegans* researchers to even think about using these models at all? Single cell tasks rely heavily on gene space definition and transformation, so it is not an apples-to-apples comparison across models if the gene space is different in each one. Additionally, the best-performing model (of the three, two using partial gene space) is the only one that used pretrained protein embeddings – but there are numerous other differences between the models. Can we really conclude that “protein-based gene representations are the most robust cross-species strategy”?

The benchmark itself also has some issues. It would be good to have some in-distribution task (generalization to a close species?) to serve as baseline, to make it easier to interpret OOD performance. Also: (1) fig 1d really shows that there is class imbalance in real time data distribution

that could have driven the correlation; (2) visually, scGPT and geneformer failed the task in 1d but somehow still achieve > 0.8 spearman correlations; and (3) it would have been good to show the ground truth annotations used to evaluated performance.